



Uganda National Roads Authority

NATIONAL ROADS SITUATIONAL

ANALYSIS REPORT

Uganda National Roads Authority

Network Planning Department

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Executive Summary

The directorate of Network planning and Engineering is mandated to; Collect and update road network and bridge asset data, Maintenance of Road, and Bridge data bank and Road Management Systems; Formulate short and long term bridge and road investment plans; Collect road condition and traffic data to support development and maintenance programming of the network among other duties. The Network planning department executes these particular duties.

In that regard, annual data collection studies of inventory and condition assessment were undertaken. The principle objective of this study was to prepare a situational assessment report that reports about the condition and usage of the road network for the FY 2019/20.

A total of 2817km of paved roads and 12,823km of unpaved roads were assessed. Accordingly, the average condition of the paved roads based on Roughness measurements was classified as “good” and unpaved roads based on the visual condition ratings as “fair”. Average IRI was found to be 3.16m/km with 450km found to be in the poor zone. These shall require immediate intervention.

The percentage of unpaved roads in poor and very poor category has remained at 23% like in the previous assessment of 2018.

The traffic data of the UNRA network is high with 28% of the paved roads carrying more than 5,000 vehicles per day, 11% carrying over 10,000 vehicle per day and 72% of the unpaved roads carrying more than 500 vehicles per day. When the AADT is disaggregated, it is clear that most dominant vehicle class travelling on the road network is motorcycles or scooters (boda-boda).

1. Introduction

This document provides results of the situational analysis, which is an integral part of the Uganda National Roads Authority (UNRA), planning cycle. It provides an understanding of the composition of UNRA's road network, traffic information as well as the current condition. A precise understanding of the current condition of a network is the first major step in infrastructure planning. Therefore, the findings herein are usually inputs in the needs analysis whose outputs are; preventive maintenance, rehabilitation and upgrading needs of UNRA road network, also referred to as an Investment Plan or National Road Investment Plan (NRIP).

The purpose of this analysis was to report on the assets and further prepare inputs into the needs analysis for the road network. Due to the access restrictions to office at the critical stage of the study, it was not possible to do the analysis for this report using the dTIMS software customised with the World Bank's Highway Development and Management (HDM-4) pavement performance and prediction algorithms. Therefore, the analysis for this FY 2019/20 was done outside the system using programmable spreadsheets.

2. The UNRA Road Network

Summary:

- UNRA has 21,010 km of road under its jurisdiction.
- 26% (5,398 km) is paved, and 74% (15,612 km) unpaved.
- 2,817 km (52%) of paved roads was analysed in this report.
- 12,823 km (82%) unpaved roads was analysed.

Network Classification and Length

The total length of the national road network is 21,010 km. Of these 5,398 km are paved roads and 15,612 km are unpaved roads. Furthermore, according to functional classification, 12.3% of this network is Class A, 13.5% Class B, 73.7% is Class C and 0.4% is Class M as indicated below.

Functional Classification	Bituminous, km	Unsealed, km	Total, km	% Total	Description
A	2594	0	2594	12.3%	International Trunk Roads
B	1395	1447	2842	13.5%	National Trunk Roads
C	1323	14165	15489	73.7%	Primary Roads
M	85	0	85	0.4%	Expressways
Total	5398	15612	21010		

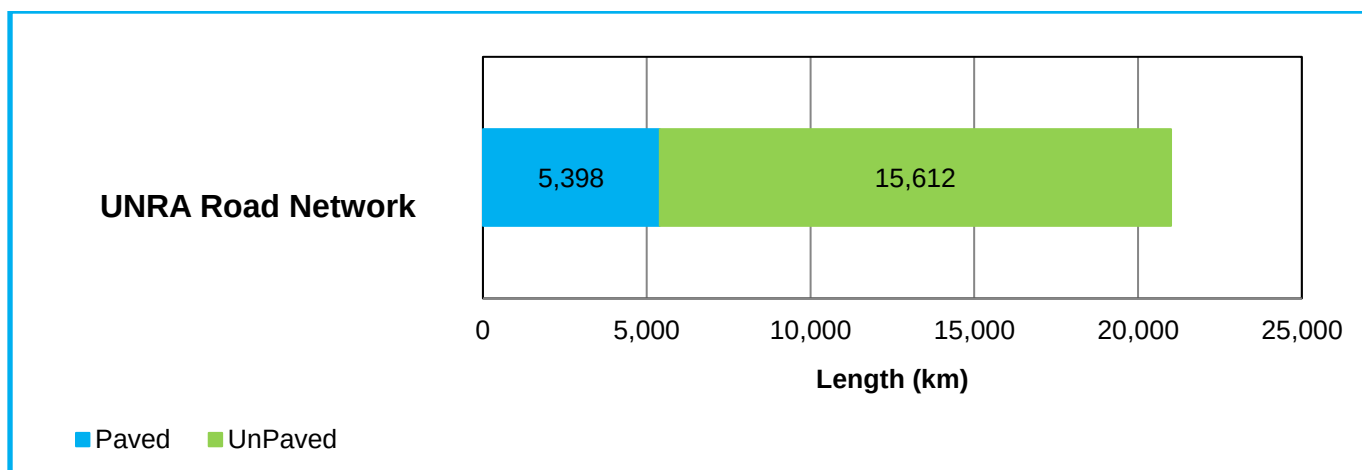


Figure 3.1: Split between Paved (26%) and Unpaved (74%) Road lengths of UNRA Network (2020)

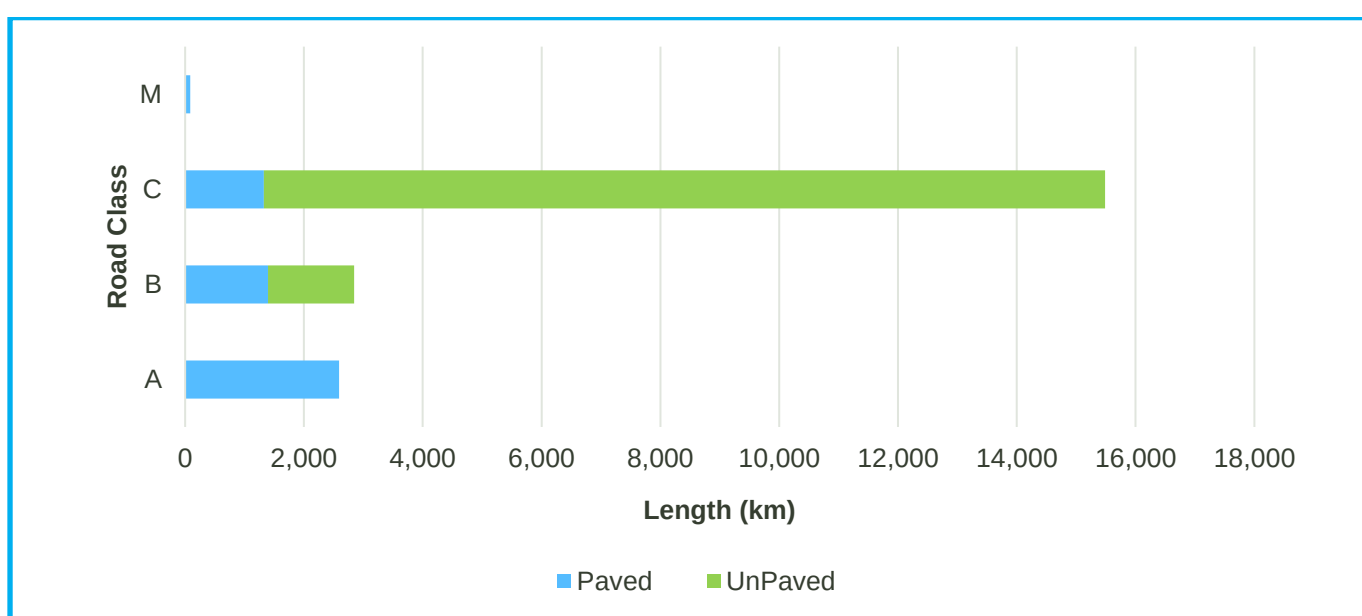


Figure 3.2: Functional Classification of UNRA Road Network.

The national network has increased from 20856km to 21010km in total lane length. This was discovered following the verification in the stations of Moroto, Kotido, Kitgum, Moyo and Luwero in 2019, as well as the continuous feedback from stations. There were also short access links that were not in the database. As well as roads that were not fully motorable at the time the network was being mapped but have now been connected. The table below shows the changes;

Road Name	Length	Type	Station	Comment	Change in length
Kikondo-Kikandwa-Nvuye	7.4	Unsealed	Luwero	New on network	7.4
Mate - Kabulamuliro-Kakinga	9.2	Unsealed	Luwero	Reduced length from 18.9km to 9.2	-9.7
Sungira Hill Access	1.6	Bituminous	Luwero	New on network	1.6
Ngoma - Kasozi - Kyamukonda	42	Unsealed	Luwero	Created Link to Kyamukonda	42
Nakitooma Access	2.4	Unsealed	Luwero	New on network	2.4
Kahunga - Nyakyera-Rukoni	31	Unsealed	Mbarara	Road name and length revised from 25.1 to 31km	5.9

Nyanga Swamp	0.7	Unsealed	Mbarara	New on network	0.7
Kityaza Bugango access	1	Unsealed	Mbarara	New on network	1
Bubandi - Kahuka -Malindi	11	Unsealed	Fort Portal	Edited length from 7 to 11km	4
Access to Muhooti Army Barracks	6	Unsealed	Fort Portal	Edited length from 4.4 to 6km	1.6
Kyehooro Link 3	6.7	Unsealed	Hoima	Adjusted legnth from 12.5km to 6.7	-5.8
Isimba-Ntoma	14	Unsealed	Masindi	Adjusted length from 6.5 to 14.	20.5
Kaliro railway access	3	Unsealed	Jinja	Added 2km to length	2
Koputh Junction-Orom	60	Unsealed	Kotido	Edited length from 12.8 to 60km	47.2
Koputh Junction - Lolelia	22	Unsealed	Kotido	Edited length from 7.7 to 22km	14.3
Lwampanga - Lwabyata	32.0	Unsealed	Luwero	Corrected	
Ngoma-Lugogo	7.5	Unsealed	Luwero	Deleted because of repetition	-7.5
Kakinga-Mate-Kabulamuliro	9.0	Unsealed	Luwero	Deleted because of repetition	-9.0
Ruerere - Biraara - Kanungu	24.7	Unsealed	Kabale	New on Network	24.7
Kituti-Buhuka	10.1	Bituminous	Hoima	New on Network	10.1
				Total change in length =	153.4

Note also that the bituminous surfacing reported now is as per substantial completion of roads. There was an error carried forward in the previous reporting where km of tarmac were added based on equivalent km in relation to progress reports. This has since been revised to report roads handed over as substantially complete.

Changes in Length (2008 – 2020)

Category/ Year	08/9	09/10	10/11	11/12	12/13	13/14	14/15	15/16	16/17	17/18	18/19	19/20
Paved (Km)	2916	3119	3264	3317	3490	3795	3919	4157	4257	4636	5015	5398
Unpaved (Km)	7137	7311	17120	17683	17510	16749	16625	16387	16287	15908	15841	15612
Total	10053	10430	20384	21000	21000	20544	20544	20544	20544	20544	20856	21010

The paved road network is now 26% of the total network.

Bridges and Structures

Bridges and structures were also verified and redefined. We categorized them into Bridges and Major Culverts as the two have different maintenance needs and life spans. We now have a total of 672No. Validation of this data has been on going and the numbers are likely to change due to construction of new bridges and upgrading of roads. Inventory of some of the roads under upgrade is not yet covered.

Bridges	379No
Major Culvert Structures	293No

The Bridge condition status report is attached.

Length Investigated

The total length of the national road network investigated in this analysis is based on the length of road with complete visual assessment information in 2019/2020. In total 15,640 km of road is investigated of which 2,817 km are paved roads and 12,823 km are unpaved roads. This is 74% of the total road network, being 52% of the paved roads and 82% of the unpaved road network.

Both unpaved and paved roads data collection was done using in-house capacity. Road condition surveys using the Mobicap tool were done for both unpaved and paved roads and IRI surveys using the ROMDAS system.

Reasons for incomplete data collected was due to the ongoing works on some of the roads at the time. A total of about 1600km of upgrading works and 600km of rehabilitations were on going. Data collection was also affected by the COVID 19 pandemic, as the team could not travel upcountry.

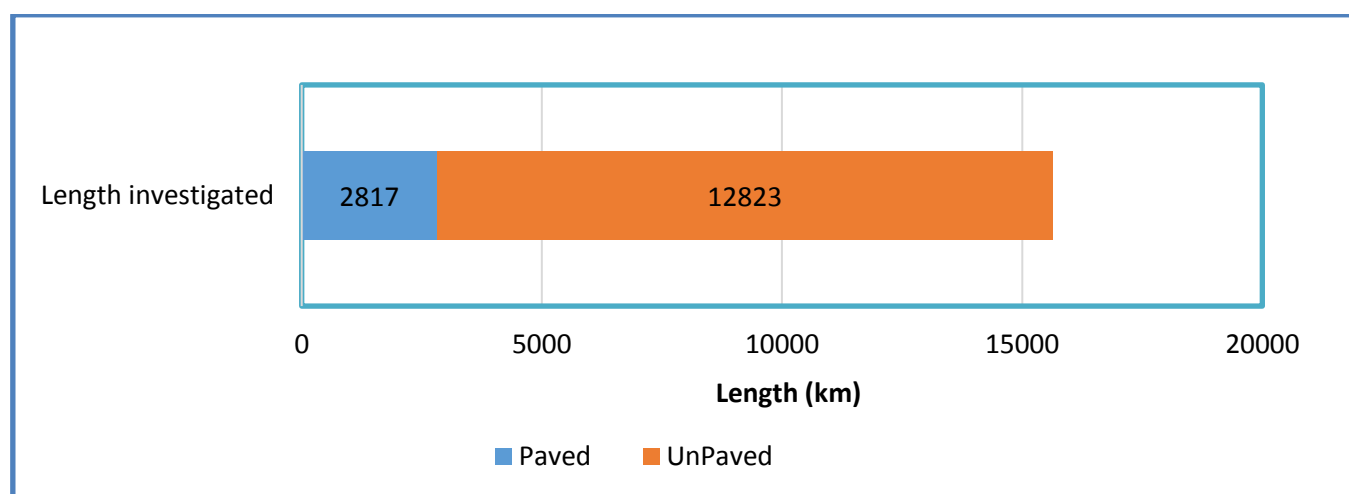


Figure 3.3: Paved and Unpaved road lengths investigated (km), 2020

3. Traffic Information

Summary:

Approximately, 15,633 million vehicle-kilometres are travelled annually on the UNRA road network.

Approximately 9,415 million vehicle-kilometres are travelled annually on paved roads and approximately 6,217 million vehicle-kilometres annually on unpaved roads.

Approximately 28% of the paved network carries more than 5,000 vehicles per day and 72% of unpaved roads carries more than 500 vehicles per day.

Traffic analysis was done using the traffic data collected by UNRA staff and Data Collection Consultants in the second and third quarter of 2018/2019 at 310 manual counting stations. This data was then projected to 2020 using traffic growth factors derived from analysis of data collected from

the Automatic Traffic Counters (ATCs). Traffic information is expressed as the Annual Average Daily Traffic (AADT), travelling on each road link of the road network. Furthermore, the product of the AADT on a road link and its length represents the daily vehicle-kilometres travelled on that link in one day. Summarised for the road network, they represent the total distance travelled by vehicles on the road network on a daily, or yearly basis. Approximately, 15,633 million vehicle-kilometres are travelled each year on the national road network. Approximately 9,415 million vehicle-kilometres are travelled annually on paved roads and approximately 6,217 million vehicle-kilometres annually on unpaved roads.

Figure 3-1 shows that although the investigated paved road network is 26% of the total road network investigated, it carries 60% of the network's traffic. A focus on the maintenance and rehabilitation of paved roads will therefore influence more road users per kilometre than unpaved roads.

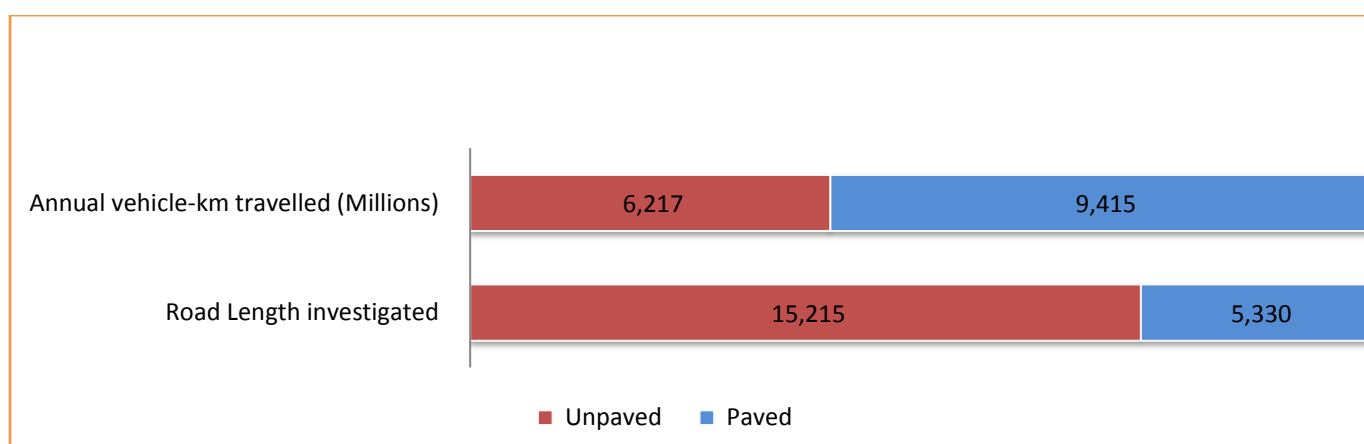


Figure 3-1: Network Length investigated Vs vehicle-kilometres travelled, 2020

3.1 Traffic per road type

The distribution of traffic volumes for paved and unpaved roads is given in Figure 3-2 and Figure 3-3 respectively. These figures show the portions of network carrying low to high categories of traffic. Approximately 72% of the paved network carries less than 5,000 vehicles per day while 17% carries 5,000 to 10,000 vehicles per day and the other 11% carries more than 10,000 vehicles per day. For unpaved roads, approximately 28% of the unpaved network carries less than 500 vehicles per day while 30% carries 500 to 1,000 vehicles per day, 40% carries 1,000 to 5,000 vehicles per day and the other 2% carries more than 5,000 vehicles per day.

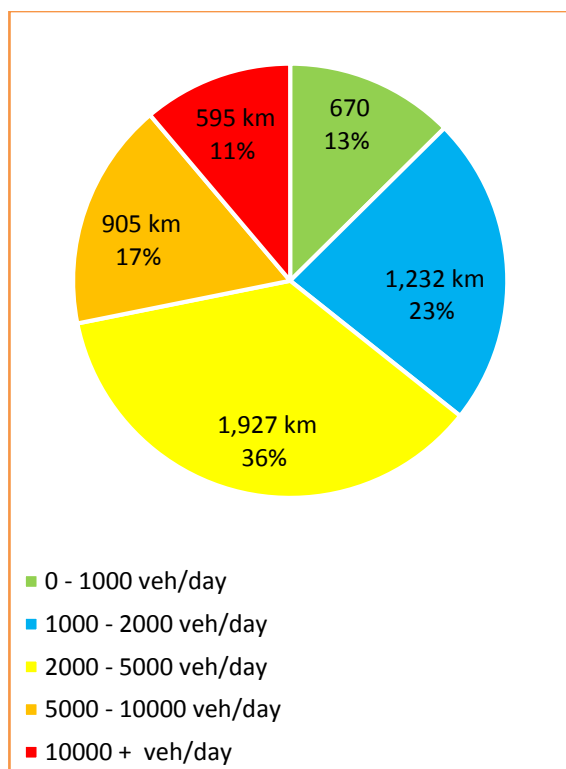


Figure 3-2: Traffic on paved roads, 2020

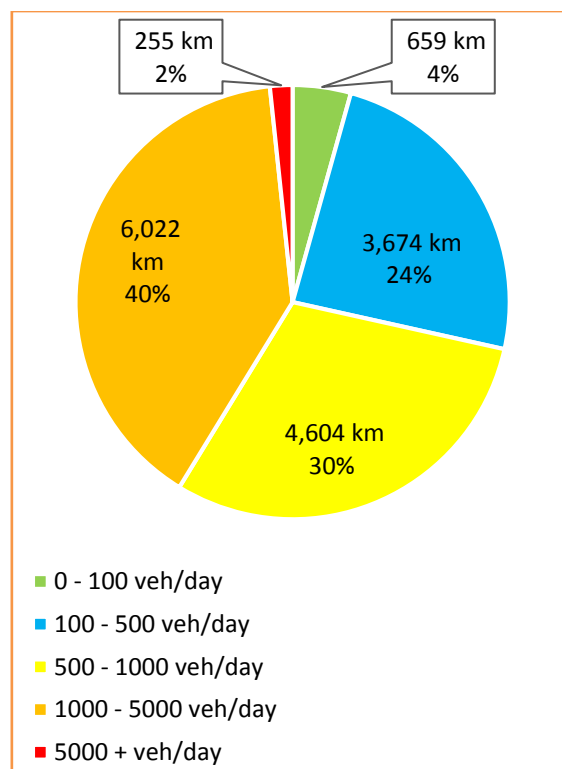


Figure 3-3: Traffic on unpaved roads, 2020

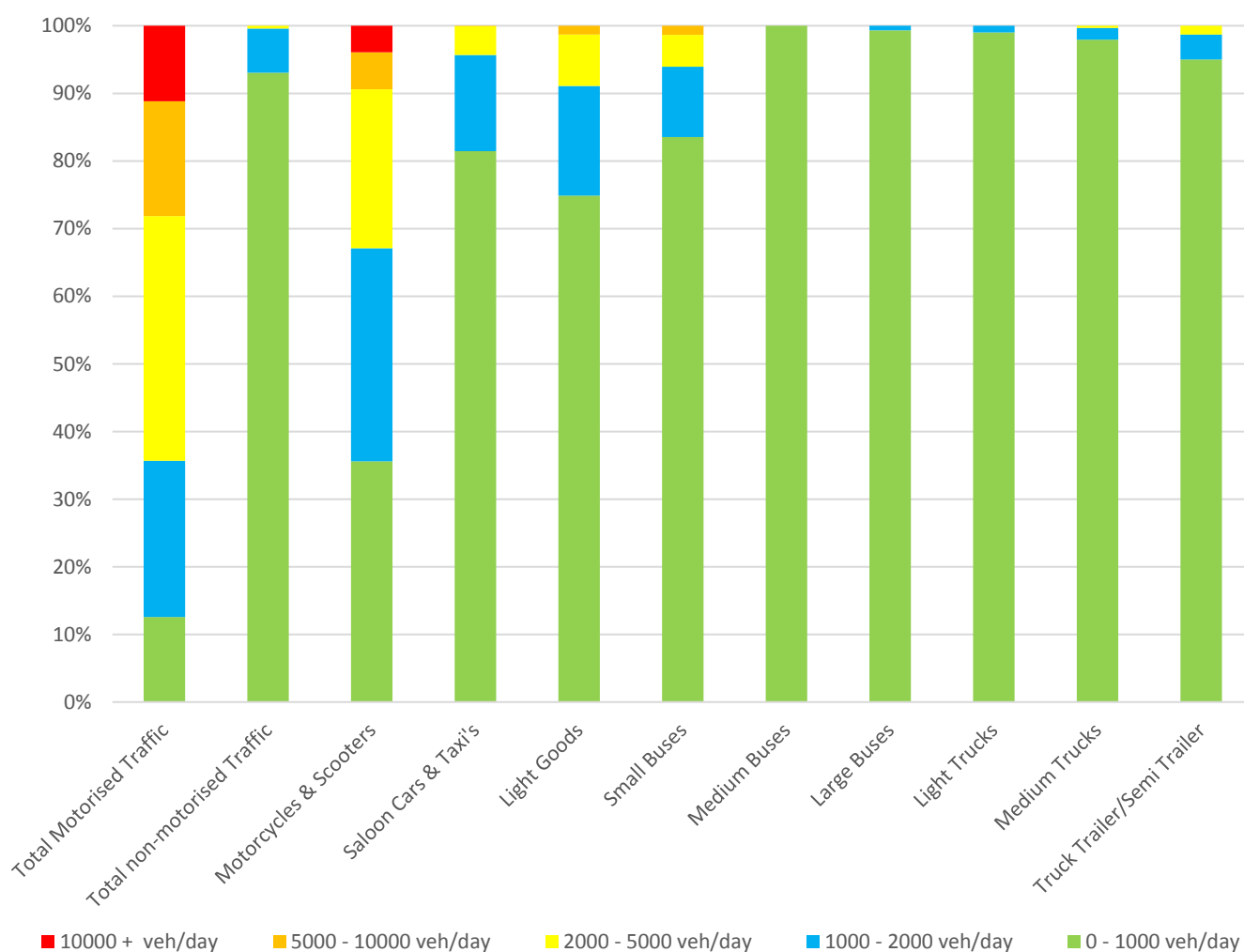


Figure 3-4: Traffic distribution on paved roads per Vehicle Type, 2020

This figure shows the vehicle classes carrying low to high categories of traffic on the paved road network.

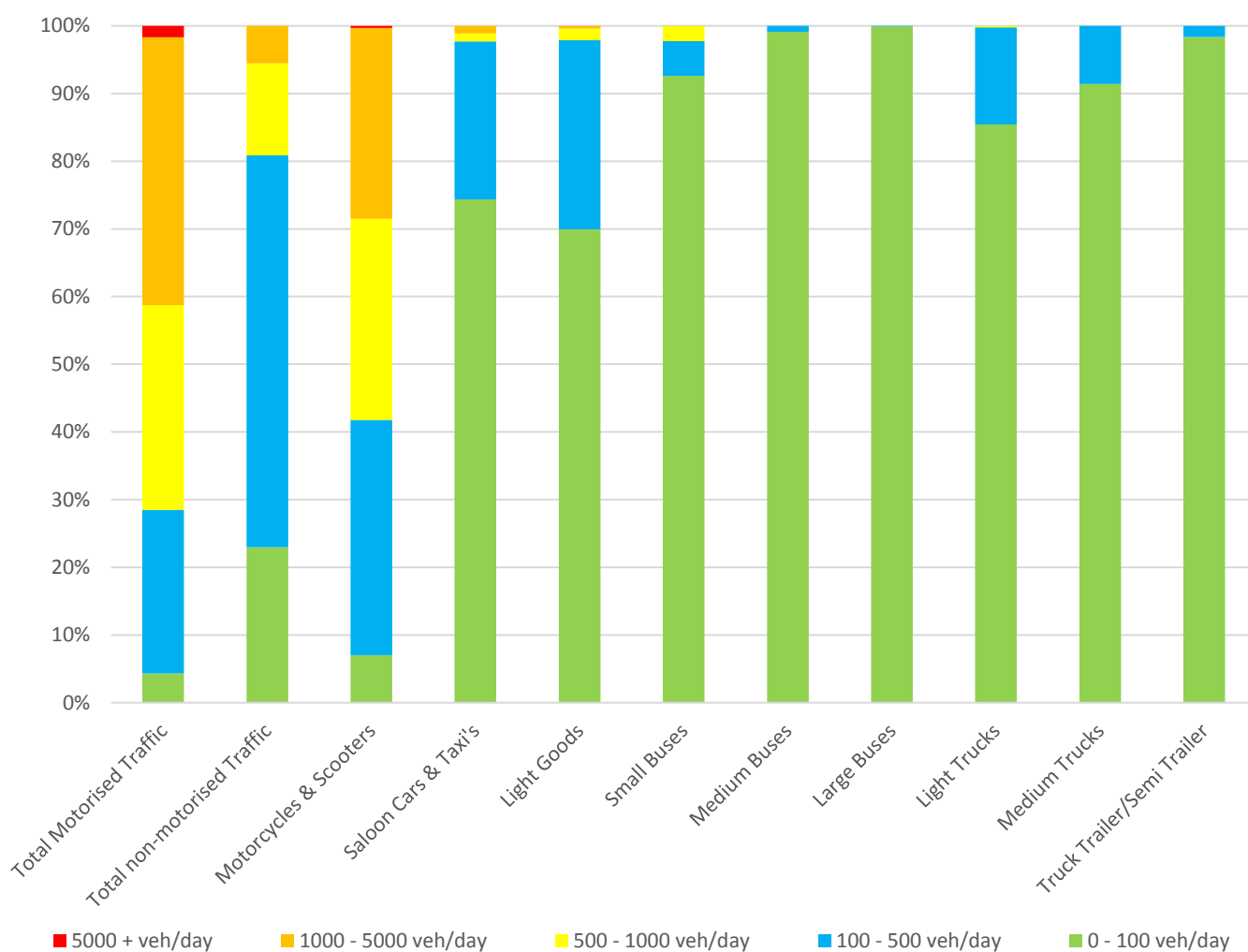


Figure 3-5: Traffic distribution on unpaved roads per Vehicle Type, 2020

This figure shows the vehicle classes carrying low to high categories of traffic on the Unpaved road network.

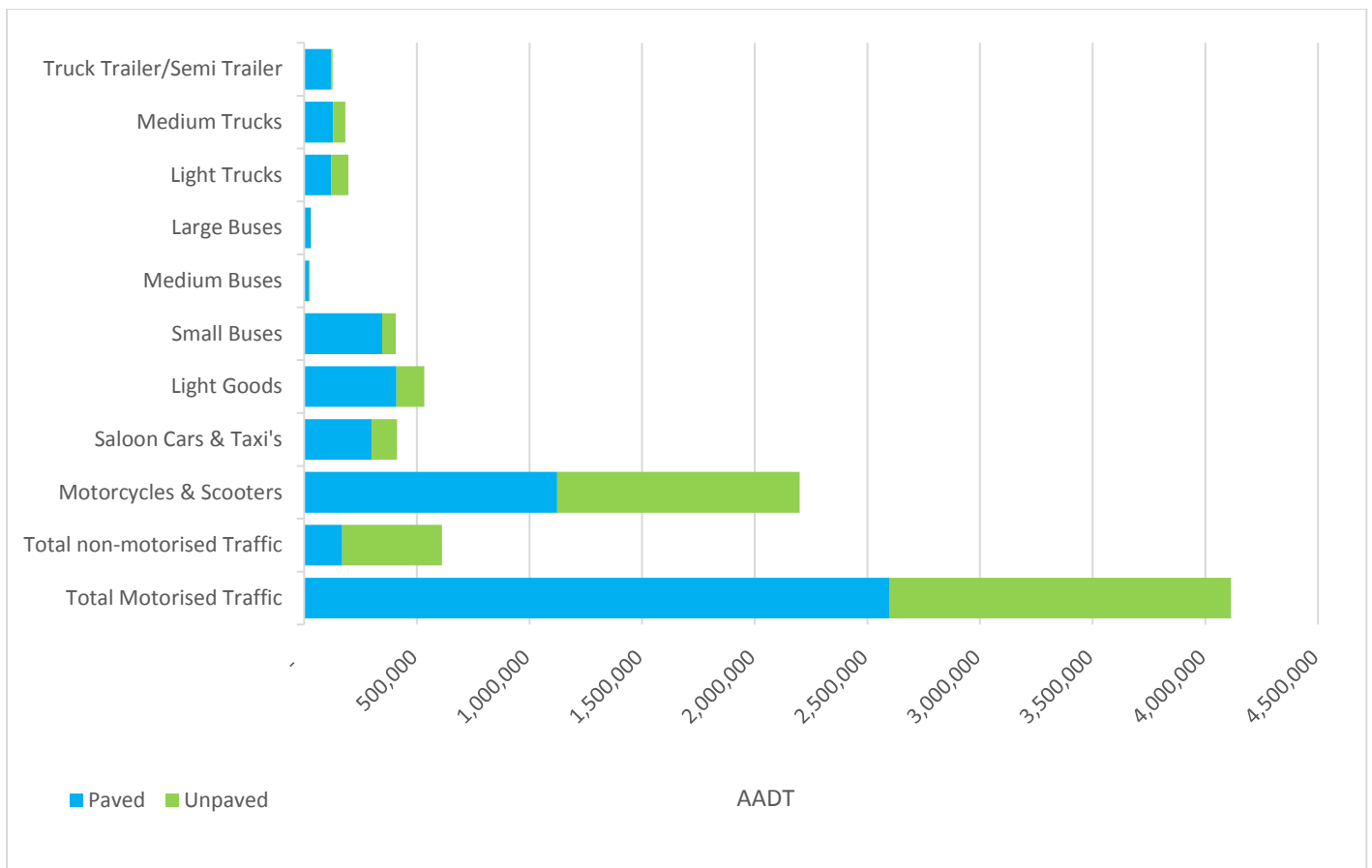


Figure 3-6: AADT Traffic distribution per Vehicle Type, 2020

Figure 3-4 indicates the AADT vehicle split on the paved and unpaved road network. It shows that the paved road network carries the highest traffic volumes. Generally, Motorcycles & Scooters (Boda Bodas) have the highest traffic volume on the entire network followed by Light Goods, Saloon Cars & Taxi's and Small Buses and Medium buses, Large Buses and Truck Trailers/Semi trailers have the least volumes.

4. Paved Roads Network Condition

Summary:

- *Moderate to severe 'All Cracking' is noted on 31% of the paved.*
- *31% of roads have severe to moderate ravelling (surface disintegration) and immediate maintenance is required on these roads.*
- *There is severe to warning roughness on 16% of the road network.*
- *28% of the road network exhibits rutting in moderate to severe state.*

4.1 Background to the VCI

The assessment data, expressing the carriageway condition through the assessment ratings of pavement defects are utilised to calculate a single Visual Condition Index (VCI). This VCI is

calculated for each paved road segment (typically of 1 km length) that was surveyed. VCI is a percentage index ranging between 0 and 100; 0 represents a road segment in very poor condition and 100 represents a road segment in very good condition. During this analysis the VCI is used as follows:

- An average network VCI is calculated representing the average condition of the paved road network or any sub-network (e.g. regional network). This average network VCI is weighted by length and is used to describe/investigate the average condition of the road network/sub-network.
- The VCI is furthermore used to categorise the condition of a paved road segment, the paved road network or sub-network into one of five condition categories. The condition categories adopted are:
 - Very Good = 86% to 100%
 - Good = 71% to 85%
 - Fair = 51% to 70%
 - Poor = 31% to 50%
 - Very Poor = 0% to 30%

4.2 Visual Assessment and Distress Graphs for Paved Road network

The condition of flexible pavements was assessed by visual inspection and by profile measurement equipment. UNRA staff executed paved visual assessments and paved roughness measurements in 2020. The Pavement Management System (PMS) of the UNRA RMS was updated with this data following basic data validation.

Figure 5-1 is a schematic representation of the distress items showing the total percentage of paved roads rated as 1  (very good or non-existent), 2  (slight, good or isolated occurrence), 3  (discernible, fair or moderate occurrence), 4  (warning, poor or general occurrence) and 5  (severe, very poor or extensive occurrence). These ratings can indirectly be related to the proportion of the paved road network with  none,  isolated,  moderate,  warning and  severe problems.

According to international standards, a “healthy” road network has less than 10% poor and very poor roads. Attention should thus be given to the distresses and condition ratings exceeding this benchmark for warning and severe problems. According to the visual assessments and automated measurements the following conclusions can be made with regards to the paved roads of UNRA:

- Cracking
- Moderate to severe ‘All Cracking’ is noted on 31% of the paved roads with another 32% exhibiting isolated cracking and 37% exhibiting no cracking. There are moderate to severe ratings for wide cracks on 27% of the paved road network with 30% exhibiting isolated wide cracking. Wide cracking is a particular concern

because these cracks indicate the disintegration of the structural layers in the wheel paths. Wide cracks also initiate the formation of rutting. Preventive measures should be considered as soon as possible to delay further deterioration with the development of secondary distresses.

Ravelling	■ Ravelling is the loss of aggregate material from the road surfacing. In seals, this manifests through stone loss, in asphalts through disintegration of the surfacing. ■ 9% of roads have severe to warning ravelling and immediate maintenance is required on these roads. Future maintenance of roads with moderate ravelling is crucial since it accounts for 22% of roads which could easily deteriorate to warning condition.
Potholes	■ 5% of roads are in a moderate to severe condition due to the occurrence of potholes. Isolated potholes occur on a further 23% of the network.
Roughness (measured)	■ From IRI surveys, 16% of the road network has severe to warning roughness. These roads typically require some form of rehabilitation. ■ 32% of the road network is in a moderate condition regarding roughness. Although some of these roads might not require immediate attention it is crucial to monitor the condition continuously and apply the right treatment at the right time to delay further deterioration.
Rutting	■ 28% of the road network exhibits rutting in moderate to severe state and 61% with isolated rutting.
Width Loss	■ There is very minimal width loss (due to edge break) on the network whether shoulders exist or not with 98% of the network exhibiting no edge break.
Edge Drops	■ There is also very minimal edge drop on the network with 98% of the network with no edge drop.
Bleeding	■ 3% of the paved network exhibits bleeding in severe to warning states with 44% having isolated bleeding.
Shoulder condition	■ The 2m width is available to the road user as emergency stopping area. There are isolated problems on 23% of the road network and 9% in moderate to severe condition.
Drainage on road	■ This refers to surface drainage and ability of road to drain water from the surfacing. Generally not a problem with only 22% in moderate condition and 5% in severe condition.

Drainage off road ■ Referring to condition of side drainage, mitre drains and cross drainage underneath the road. 5% of the paved road network in a warning to severe conditions, 73% in warning state and 21% with isolated drainage problems.

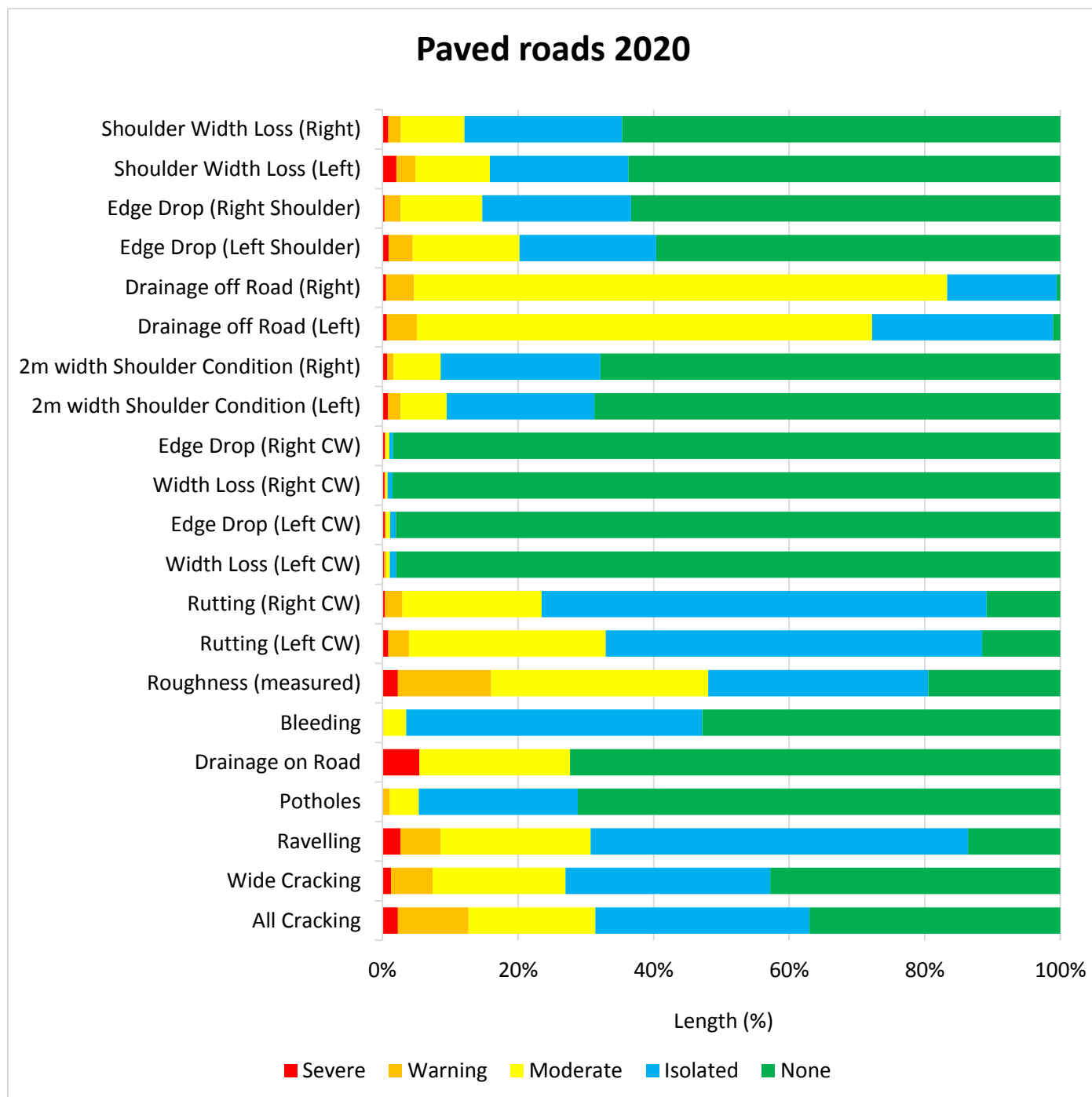
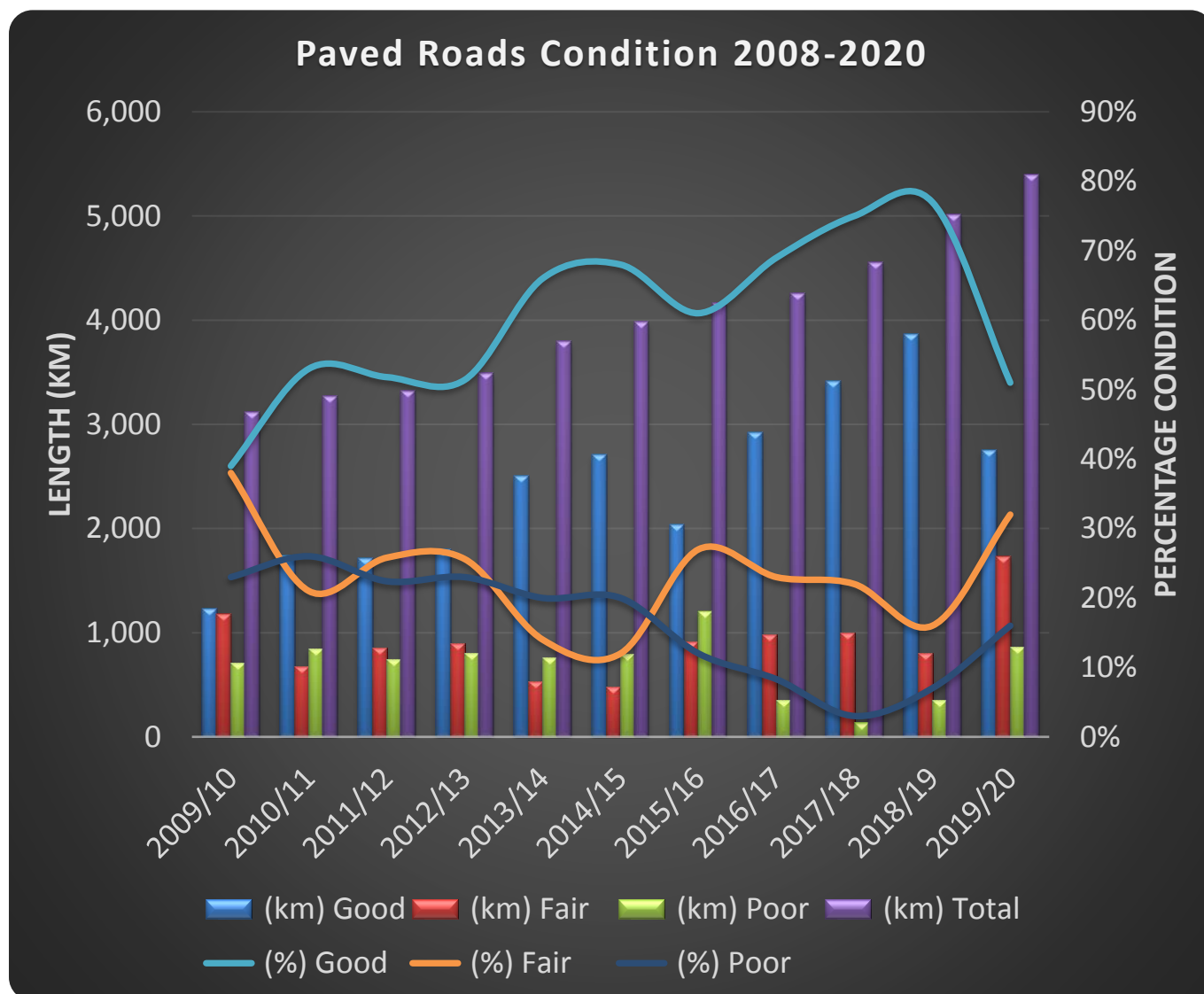


Figure 5-1: Measured values and distress ratings of paved roads, 2020

In summary, the paved network is in good condition. From a pavement management perspective, roads appear to be structurally sound, but cracking, roughness and rutting indicate some structural deterioration (rated as severe to warning). A preventive maintenance regime of timely patching, crack sealing, diluted emulsions and resurfacings will contribute significantly to delay the structural deterioration by preventing water from entering the pavement structures.

4.3 Condition trend for paved roads



The figure above shows the condition trend for the paved roads from 2009/10 to 2020.

TABLE 1: CONDITION OF ROADS IN KM							
Year	(km)				(%)		
	Good	Fair	Poor	Total	Good	Fair	Poor
2009/10	1,230	1,180	709	3119	39%	38%	23%
2010/11	1,742	680	843	3264	53%	21%	26%
2011/12	1717	856	744	3317	52%	26%	22%
2012/13	1,794	893	803	3490	51%	26%	23%
2013/14	2505	531	759	3795	66%	14%	20%
2014/15	2,707	478	796	3,981	68%	12%	20%
2015/16	2040	913	1204	4,157	61%	27%	12%
2016/17	2924	979	354	4,257	69%	23%	8%
2017/18	3413	1001	136	4551	75%	22%	3%
2018/19	3862	802	351	5015	77%	16%	7%
2019/20	2753	1727	864	5398	51%	32%	16%

5. Unpaved Road Network Condition

Summary:

- *The average unpaved road VCI based on the 2020 data is 63% which categorises the road network in the “fair” condition category.*
- *The gravel thickness ratings indicate the need for urgent re-gravelling on 15%. 3% has no gravel at all and 12% has gravel less than 50mm.*
- *The roughness ratings on the unpaved network indicate severe to moderate condition states on 35% of the road network.*
- *Roads with rutting in a moderate to severe condition accounts for almost 46% of the network.*
- *37% of all roads have severe to moderate corrugations.*
- *Erosion is very common with almost 57% of roads showing the occurrence of severe to moderate erosion.*
- *Drainage condition and formation level is poor and very poor on 13% of the network.*
- *% of Roads in very poor and poor condition has decreased from 25% in 2018 to 23% in 2020.*

5.1 Background to Unpaved VCI

The assessment ratings are utilised to calculate a Visual Condition Index (VCI) similar to the VCI of paved roads. This unpaved VCI is calculated for each unpaved road segment that was surveyed. VCI is a percentage index ranging between 0 and 100; 0 represents a road segment in very poor condition and 100 represents a road segment in very good condition. During this analysis the VCI is used as follows:

- An average unpaved network VCI is calculated representing the average condition of the unpaved road network or any sub-network. This average network VCI is weighted by length and is used to describe/investigate the average condition of the road network/sub-network.
- The VCI is furthermore used to categorise the condition of an unpaved road segment, the unpaved road network or sub-network into one of five condition categories. The condition categories adopted are:
 - Very Good = 86% to 100%
 - Good = 71% to 85%
 - Fair = 51% to 70%
 - Poor = 31% to 50%
 - Very Poor = 0% to 30%

5.2 Visual assessment of Unpaved Road network

Unpaved visual assessment data was collected in 2020 and the PMS updated with these data.

The assessment data expresses the condition and characteristics of the wearing course material and the functional condition of the gravel roads as a set of rating scores, rating these as 1 ■ (very good or non-existent), 2 ■ (slight, good or isolated occurrence), 3 ■ (discernible, fair or moderate occurrence), 4 ■ (warning, poor or general occurrence) and 5 ■ (severe, very poor or extensive occurrence).

These ratings can indirectly be related to the proportion of the unpaved road network with ■ none, ■ isolated, ■ moderate, ■ warning and ■ severe problems.

According to the visual assessments, the following conclusions can be made about the unpaved roads of UNRA (refer

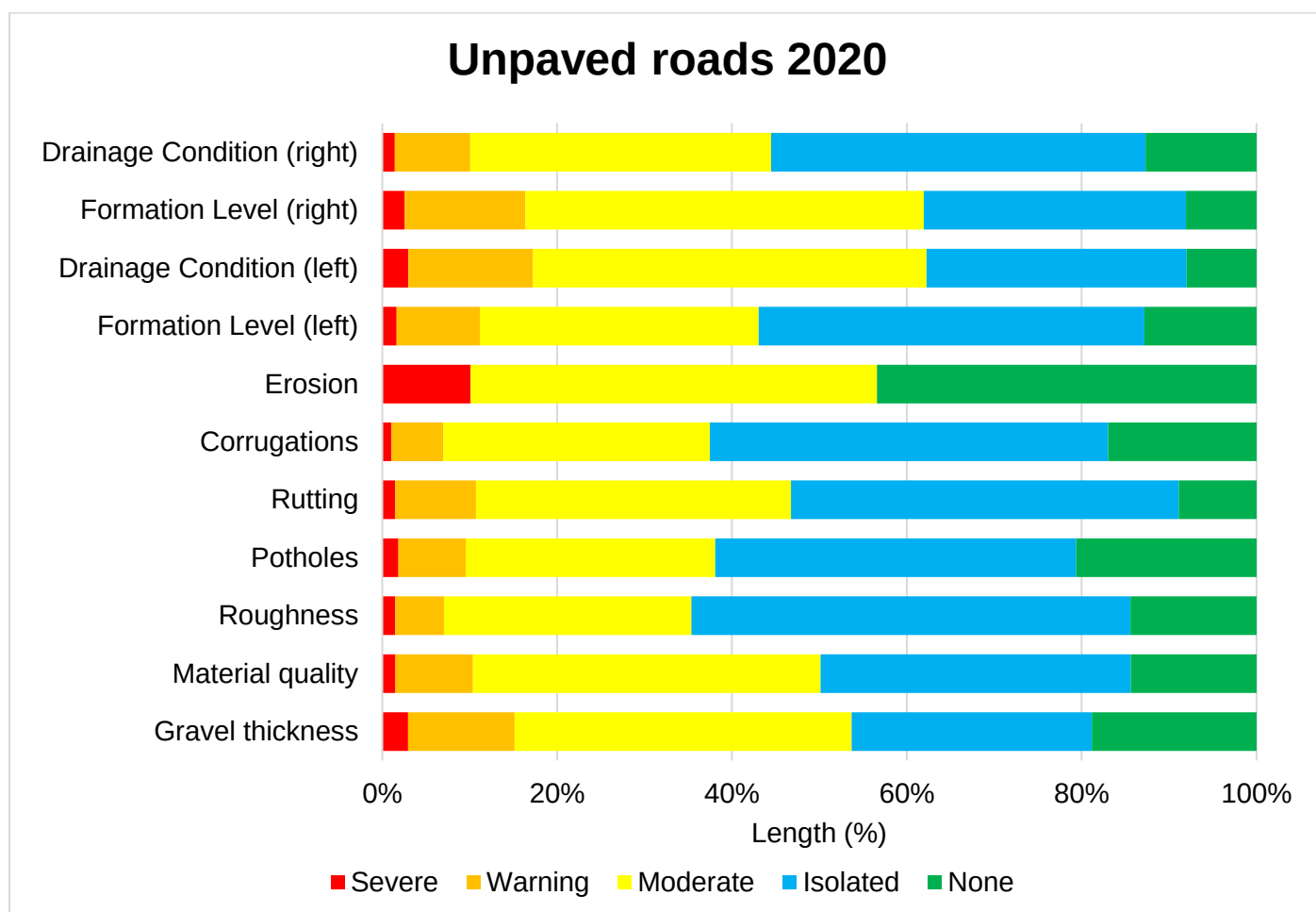


Figure 6-1):

- Gravel thickness
 - The gravel thickness ratings indicate the need for re-gravelling on 15% of the unpaved road network, with 3% having no gravel, another 12% having less than 50mm gravel wearing course and severe exposure of the sub grade. A further 39% is in the moderate

Material quality	condition state, with subgrade exposed in local areas (average of 50 to 100mm gravel thickness). ■ 2% of roads having severe material quality exhibiting poorly distributed range of particle sizes and/or zero or excessive plasticity, 9% of roads was rated with warning material quality (poor particle size distribution). A further 40% exhibits cracking, loose material or stones clearly visible, thus rated as material quality of moderate state.
Roughness	■ The roughness ratings on the unpaved network indicate 35% of the road network to be warning to moderate condition. This will result in high road user costs.
Potholes	■ The high occurrence of potholes especially in severe to moderate states (39%) and 41% in isolated state is indicative of inadequate maintenance.
Rutting	■ Roads with rutting in a moderate to severe condition accounts closely to 46% of the network. These roads are in need of blading and re-graveling.
Corrugations	■ 37% of all roads have severe to moderate corrugations. A further 44% has isolated corrugations.
Erosion	■ Erosion was rated serious on 10% of road network. There is a need to repair erosion more regularly.
Drain condition and formation level	■ Drainage condition and formation level indicate the ability of roads to drain water away, transversely underneath and longitudinally next to them. Roads with severe or warning formation levels, would act as a canal, this is the case for 14% of the unpaved road network. 39% of the unpaved roads have moderate formation levels indicating that the drainage is level with ground level. ■ Furthermore, the drainage condition is warning to severe for 13% the network with moderate drainage conditions existing on 40% of the network.

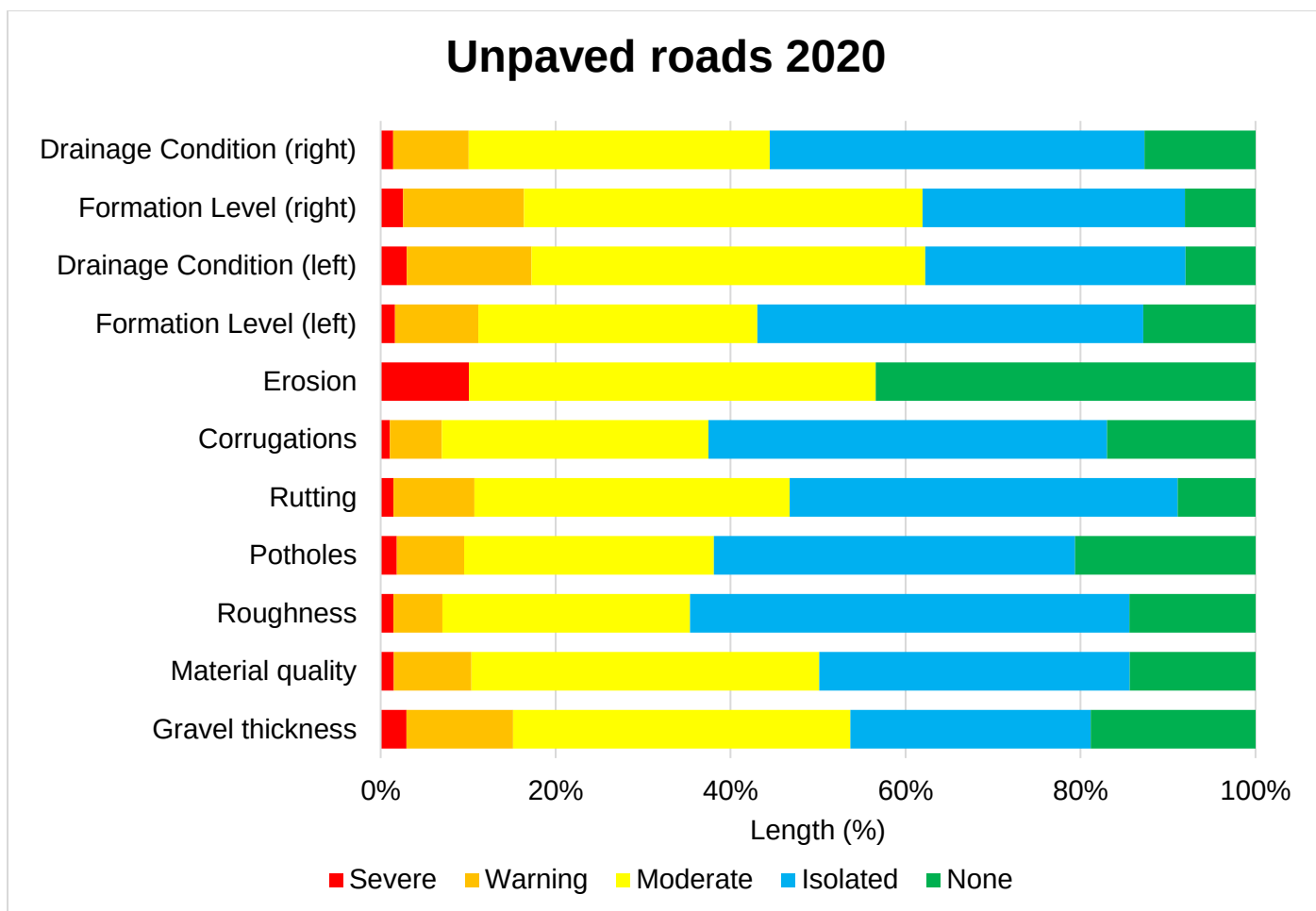


Figure 6-1: Distress, characteristic and condition ratings of unpaved roads, 2020

In summary, the unpaved network is in fair condition. From a road user’s perspective, material quality and possibly non-sufficient maintenance regime contribute to the development of rough roads with potholes, rutting and corrugations. From an unpaved roads management perspective, gravel thickness problems, low formation levels and poor drainage conditions result in erosion, wetting of the gravel layers and subgrade which result in potholes and rutting. A well planned gravelling programme should be supported by an effective maintenance regime of blading (to maintain cambers and limit roughness), cleaning of side drains, mitre drains and cross drainage.

5.3 Average VCI for Unpaved roads

The average unpaved road VCI based on the 2020 data is 63% which categorises the road network in the “fair” condition category, this is 2% improvement as compared to the average VCI calculated on the 2018 visual assessment data.

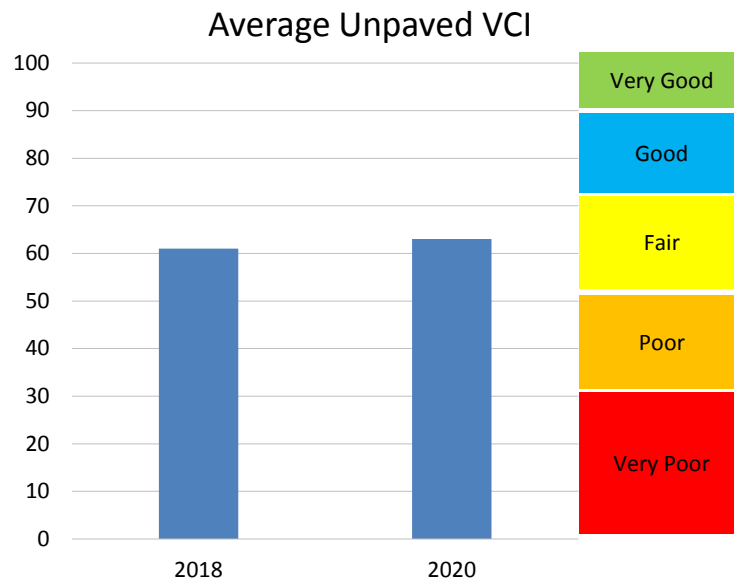


Figure 6-2: Average Unpaved VCI comparison

5.4 Condition distribution graphs for the VCI of Unpaved roads

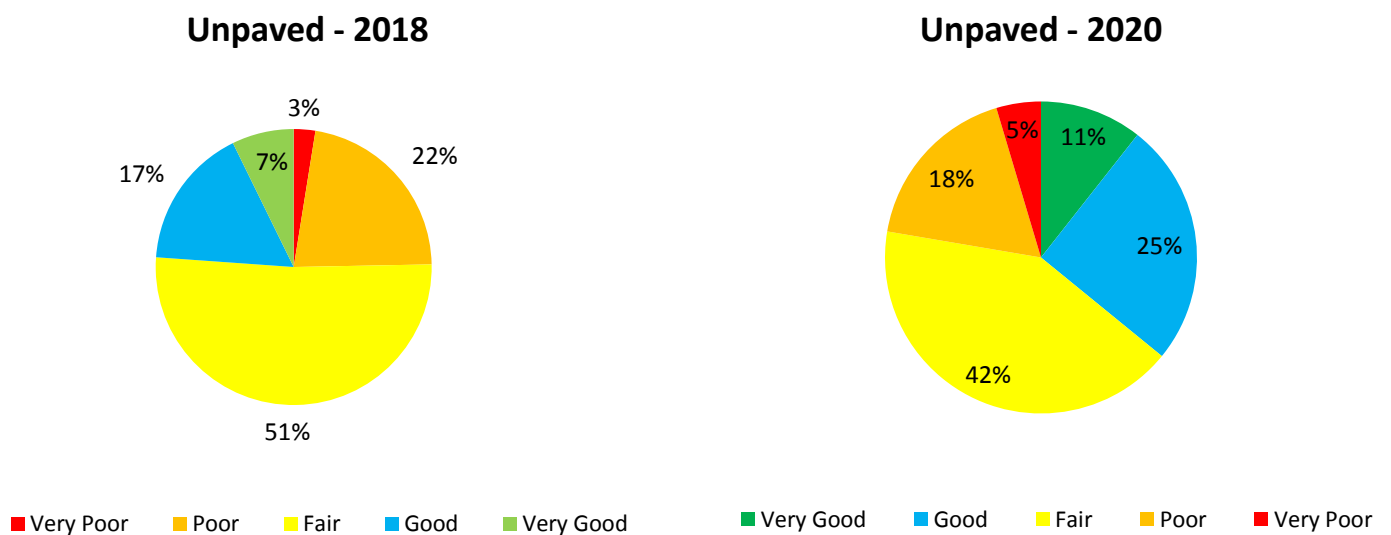
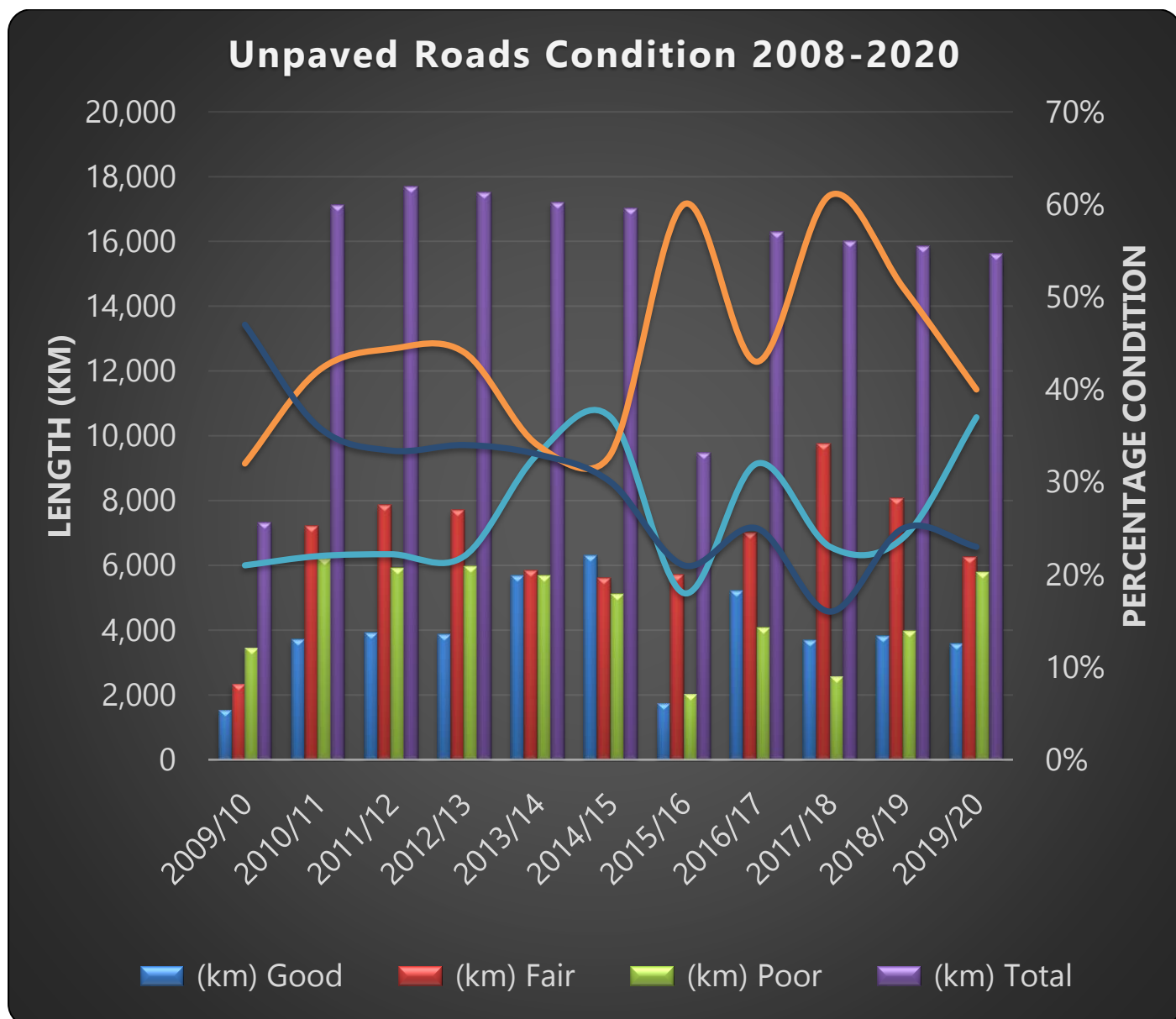


Figure 6-3:Unpaved VCI distribution for 2018

Figure 5-4:Unpaved VCI distribution for 2020

The figures show that the percentage of roads in very poor and poor condition has reduced from 25% in 2018 to 23% in 2020. The roads in fair condition has decreased from 51% in 2018 to 42% in 2020. The condition of roads in good and very good condition has increased from 24% in 2018 to 36% in 2020.

5.5 Condition trends for unpaved roads



The figure above shows the condition trend for the unpaved roads from 2009/10 to 2020.

	Unpaved Roads Condition (km)				Unpaved Roads Condition (%)		
	Good	Fair	Poor	Total	Good	Fair	Poor
2009/10	1,535	2,340	3,436	7,311	21%	32%	47%
2010/11	3,719	7,215	6,186	17,120	22%	42%	36%
2011/12	3,926	7,853	5,904	17,683	22%	44%	33%
2012/13	3,852	7,705	5,954	17,510	22%	44%	34%
2013/14	5,678	5,850	5,677	17,205	33%	34%	33%
2014/15	6,297	5,616	5,106	17,019	37%	33%	30%
2015/16	1,737	5,707	2,033	9,477	18%	60%	21%
2016/17	5,212	7,003	4,072	16,287	32%	43%	25%
2017/18	3,678	9,755	2,558	15,993	23%	61%	16%
2018/19	3,802	8,079	3,960	15,841	24%	51%	25%
2019/20	3,591	6,245	5,776	15,612	37%	40%	23%

6. Conclusion

The UNRA road network total length is now at 21,010km. Of these roads, 5,398km are paved roads and 15,612km are unpaved roads. Of these a new class "M" of expressways totalling to 85km.

The condition of the paved roads has been rated "good" with 52% but with a significant length that is in dire need of preventive maintenance to deter accelerated deterioration.

The high AADT on the unpaved roads indicates a growing increase in the paving more roads from gravel to tarmac.

The mode of failure of unpaved roads is mainly drainage and formation level related.

The most dominating mode of failure of paved roads is cracking with a significant portion of the northern corridor route affected all the way from Malaba/Busia to Katuna. This shall require periodic maintenance very soon.

Appendix 1: Pictures of roads on the UNRA network

Appendix 2: International Roughness Index for Paved roads

Appendix 3: Traffic Summary Spreadsheet

Appendix 4: Visual Condition Summary Unpaved roads